Project Progress Update

AgriCom, ESMT

March 13, 2026

Arjun Chahal, Jean-Christophe Scheider, Akos Peter,

Xishuang Wang, Tsung-Han Tsai, Matias Cam

Executive Summary

This update focuses on answering key questions raised from the previous report sent last week as well as proposing the next steps for project testing. The report is structured as follows: Sections 1-4 provide further information in response to questions surrounding demand proxy definition, price role, consumer segmentation, and weekly aggregation; Section 5 provides further validation of key project details with literature; and finally, Section 6 outlines the project hypotheses, data requirements, and a proposed testing strategy for review.

1. Demand Proxy Definition

Are we modeling demand for the basket as a whole or for each product individually?

We model demand at the individual product level and then aggregate into a composite index. This allows for product-specific elasticity estimation and ensures that heterogeneous demand drivers (e.g., weather sensitivity for cucumbers vs. staple behavior for tomatoes) are captured before aggregation.

What justifies the proposed weighting scheme (35/30/20/15)?

The weights are calibrated against two official German data sources. First, BMEL/BLE consumption data for WJ 2023/24 confirms tomatoes as the dominant vegetable by volume (approximately 2.3 million tonnes nationally, ~27 kg per capita), validating their leading weight. However, because these figures include processed tomato products (passata, ketchup, concentrate), the raw volume share (~67% within the basket) overstates fresh retail relevance. The 35% weight therefore reflects a deliberate discount to fresh-market consumption only.

Cucumbers rank fourth nationally at approximately 682,000 tonnes (~8 kg per capita), supporting the 20% weight. Pepper and leafy green weights are grounded in Destatis greenhouse cultivation data (2024), where peppers occupy 120 hectares and salad varieties a comparable area, justifying their combined 45% share. Leafy greens receive a slight premium over peppers (30% vs. 15%) due to their higher purchase frequency and role as a basket traffic driver.

It should be noted that Berlin-specific crop-level sales data is not publicly available. The weights therefore apply national consumption ratios with the acknowledgment that Berlin's above-average organic market penetration and restaurant sector demand may shift the relative importance of leafy greens and specialty peppers upward compared to the national baseline. Sensitivity testing across alternative weighting scenarios is planned as part of the H2 validation exercise.

Does Google Trends search interest correlate with purchasing behavior for these items?

The use of Google Trends as a demand proxy is supported by Choi and Varian (2012), who demonstrated that search query data functions as a leading indicator of consumer behavior, and by Boone et al. (2018), who found that integrating search data into grocery forecasting models reduced forecast error relative to time-series baselines. However, as Goel et al. (2010) caution, the predictive value is category-dependent and tends to be weaker for low-consideration, habitual purchases, which partially describes fresh produce. The search-to-purchase link is therefore treated as a hypothesis (H1) rather than an assumption, and will be empirically validated against AgriCom's weekly sales data once received.

2. Price Role

Will price be incorporated directly into the demand model or treated as a control variable?

Price will be incorporated directly into the demand model as a primary predictor variable. By regressing weekly demand on the organic "own-price" and comparing it to the conventional "substitute price," the model will calculate price elasticity and measure how shifts in the organic premium directly drive or suppress volume.

How will promotions or discounter pricing shocks be captured?

Promotions can be isolated using binary promotion flags (Y/N) and specific promotional pricing data which may be provided by AgriCom. These flags will act as control variables during elasticity regressions to separate regular price sensitivity from temporary promotional spikes. Additionally, capturing sales data split by retail format (e.g., discounters like Aldi/Lidl vs. full-service like Rewe) will allow the model to map discounter-specific pricing shocks directly to the "Price-Sensitive Discounters / Families" segment.

Since peppers are described as the most price-sensitive item, it would be useful to show a simple elasticity hypothesis for the basket.

We hypothesize a rank-order of own-price elasticity where bell peppers are highly elastic (expected coefficient around -1.5), cucumbers are moderately elastic (around -1.0), and the anchor staples—tomatoes and leafy greens—remain price-inelastic (between -0.5 and -0.8). To test this, we will fit a log-log OLS regression on AgriCom's weekly product-level sales volume and retail pricing data, which will allow us to extract these exact elasticity coefficients while controlling for promotional flags and seasonal shifts.

3. Consumer Segmentation

1) Premium Sustainability Buyers

- **Income Proxy**

  Central, high-income pockets (e.g., Kreuzberg, Prenzlauer Berg) help identify consumers with low price elasticity who prioritize quality, health, and sustainability coherence.

- **Organic Retail Density**

  A high concentration of bio-specialists and weekly organic markets (such as the Ökomarkt am Kollwitzplatz) identifies zones where shoppers have a high response to product narratives like traceability and wellbeing framing.

2) Price-Sensitive Discounters / Families

- **Household Size**

  Larger household sizes in mixed suburban districts help identify family-dense areas where the core motive is value optimization under strict budget constraints, as price sensitivity creates a barrier for budget-conscious segments.

- **Discounter Penetration**

  A high density of stores like Aldi, Lidl, and Rewe highlights zones with high promotion sensitivity, where shoppers shift toward private-label organics for larger planned baskets.

3) Convenience-Driven Urban

- **Single Household Share in the area**

  This can help identify "time-poor" professionals who favor pre-packed mixes and quick-commerce.

- **Delivery Service Density**

  This segment has a high tolerance for convenience premiums; the presence of "dark stores" or high delivery activity identifies these zones.

4) Community-Focused Traditionalists

- **Weekly Market Density**

  Traditionalists prefer "market-oriented neighborhoods" and independent retail. Measuring the distance from a neighborhood center to the nearest weekly market captures this "habitual purchase relationship".

- **Immigrant Density**

  Areas with high migration backgrounds (e.g., Neukölln) often feature "Turkish-market ecosystems" that provide high produce variety and serve as a hub for community-focused shopping.

5) Modern Plant-Based Urbanites (Flexitarian-forward)

- **Average Age in the region**

  Younger urban populations are the primary drivers of the "flexitarian-forward" movement and are most responsive to social media recipe trends.

- **Search Interest Index based on Google Trend**

  This segment shows high responsiveness to "themed food moments". Tracking keywords like "vegane rezepte" or "bio salat" provides a real-time demand proxy for this trend-sensitive group

4. Weekly Aggregation

The recommendation in Section 4 to additionally test daily signals (weather, events, search spikes) before finalizing weekly aggregation is methodologically prudent. Murray et al (2010)

demonstrate that weather variables have significant predictive power for produce purchasing frequency, and this effect can manifest within a week (e.g., a heat wave mid-week driving salad demand on specific days). Capturing these sub-weekly dynamics may improve forecast lead time, particularly for supply planning purposes. However, daily modeling requires higher data resolution than is likely available from public sources, making it an aspirational enhancement rather than a baseline requirement.

5. Validation with Literature

5.1 Basket Composition

The selection of a four-item fresh produce basket as a demand proxy is consistent with research on complementary purchasing behavior. Consumer panel studies in European grocery contexts consistently show that items consumed within the same meal occasion actually exhibit strong co-purchase correlations, which in turn makes basket-level demand more stable and forecastable than single-SKU demand (van Dijk et al., 2004). Additionally, as mentioned in our last update, the shorter shelf life of the selected items also justifies treating them as a high-frequency behavioral cluster, as perishability actually drives purchase regularity in a manner that is distinct from durable food categories (Hu et al., 2024).

The organic framing of the basket is supported by German market structure data. According to the Bund Ökologische Lebensmittelwirtschaft (BÖLW, 2025), Germany remains the largest organic food market in Europe by absolute sales volume. Berlin specifically exhibits above-average organic penetration relative to the national average, driven by higher education levels, urban density, and ideological alignment with sustainability (Boekhout, 2022). Furthermore, thin-skinned vegetables, like those selected in this basket, are disproportionately

purchased in organic form due to heightened pesticide concern (Hughner et al., 2007), which could validate the basket's relevance as an overall organic demand signal.

5.2 Google Trends as a Demand Proxy

The use of Google Trends data as a near-real-time demand signal is well-established in the operations and forecasting literature, though its application requires careful calibration. Choi and Varian (2012) demonstrated that search query data can predict present and near-future consumer behavior across multiple categories, functioning as a leading indicator ahead of observed sales. In the agricultural and food retail context, Boone et al. (2018) found that integrating Google Trends into sales forecasting models for grocery categories reduced forecast error relative to time-series baselines alone.

Importantly, similar to the basket weighting described in Section 1, the search-to-purchase link here requires empirical validation before being assumed. Goel et al. (2010) cautions that the predictive value of search interest can be quite category-dependent, with some stronger findings for high-consideration items (e.g., appliances, vehicles) and weaker for purchases where search precedes purchase by very little time. Thus, for our case, this creates a direct need for the validation exercise proposed later on in Section 6.

A further concern would be geographic resolution. Standard Google Trends data is available at country level, and city-level granularity (Berlin) is also achievable but comes with higher noise-to-signal ratios due to the smaller query volumes (Fleder & Shah, 2019). Weekly aggregation, as discussed, is methodologically sound here given that daily search volumes can be too volatile and susceptible to one-off events, whereas weekly windows can help to smooth that noise.

5.3 Consumer Segmentation Proxies

Area-level characteristics (income, household size, ethnic composition, organic retail density) serve as workable substitutes, as demonstrated in spatial demand modeling studies in European food retail contexts (Richards et al., 2021). However, this introduces ecological fallacy risk, the attributing of area-level patterns to individuals, and must be acknowledged as a limitation of the proxy approach.

6. Next Steps: Testing Strategy for Validation

6.1 Hypotheses for Testing

The following hypotheses are proposed for empirical evaluation at the product level:

- **H1)** Search-Sales Correlation: Weekly Google Trends indices for bio tomaten, bio salat, bio gurken, and bio paprika Berlin exhibit a statistically significant positive correlation with weekly organic sales volumes for the corresponding products in Berlin, with search interest functioning as a leading or coincident indicator.
- **H2)** Basket Composite Validity: The weighted composite demand index (35/30/20/15 weighting scheme) outperforms equal-weight and single-product benchmarks in tracking aggregate organic salad basket sales, as measured by correlation coefficient and mean absolute error against observed data.
- **H3)** Price Elasticity Differentiation: Bell peppers exhibit a higher own-price elasticity than tomatoes and cucumbers, consistent with their characterization as a premium-tier, budget-sensitive basket complement.

- **H4)** Segment Proxy Operationalization: Neighborhood-level sociodemographic variables (income, household size, organic retail density) are significant predictors of organic produce sales intensity across Berlin districts, with Premium Sustainability Buyer zones showing higher baseline demand levels and lower price sensitivity.
- **H5)** Seasonality and Weather Signal: Weekly temperature and sunshine hours exhibit a significant positive relationship with cucumber demand (and potentially pepper demand), consistent with their characterization as warm-season items with a stronger weather sensitivity in comparison to tomatoes and greens.

6.2 Data Requested from AgriCom

The following data are required from AgriCom to help enable us to perform testing of the hypotheses mentioned above. Data should be provided at weekly granularity where available, covering a minimum of 156 weeks (3 years) to ensure sufficient depth.

Data	**Variables**	**Granularity**	**Purpose**
Organic Basket Sales	Weekly unit volume and revenue for: round tomatoes, bagged leafy greens, cucumbers, bell peppers (organic SKUs, Berlin region)	Weekly, by product; district or store-cluster level if available	Ground truth for H1 and H2 (proxy calibration)
Conventional Sales (Same SKUs)	Weekly unit volume and revenue for conventional equivalents of each basket item	Weekly, by product	Required for organic premium calculation and H3 elasticity estimation
Retail Price Data	Average weekly shelf price (EUR/kg or EUR/unit) per product, organic and	Weekly, by product and retail format (discounter vs. full-service vs.	Own-price and substitute price variables; promotion indicator for H3

	conventional; promotion flags (Y/N) and promotional price where applicable	specialty)	
Distribution Channel	Sales split by retail channel: discounters (Aldi/Lidl), full-service (Rewe/Edeka), bio specialists, delivery, weekly markets	Weekly or monthly, by channel	Channel composition control & segment interaction modeling
Supply-Side Events	Flags for any possible known supply disruptions, import substitutions, or replenishment shortfalls affecting any basket item during the period that is provided	Event-level with week	In order to actually be able to isolate demand shocks from supply constraints
Historical Replenishment Data	Internal reorder frequency or distribution center throughput data for the basket items in Berlin	Weekly	If available (further validation)

6.3 Data Collected (or to be) by Our Team

Variable	**Source**	**Access Method**	**Purpose**
Search index: bio tomaten, bio salat, bio gurken, bio paprika Berlin	Google Trends	Manual export	H1 Proxy variable
Weekly temperature, precipitation, & sunshine hours	Open-meteo	API	H5 weather signal & seasonality control

(Berlin)			
Neighborhood income, household size, single-person share by district, population by age group, & migration background by district	Amt für Statistik Berlin-Brandenburg (statistik-berlin-brandenburg.de)	Open Data Portal	H4 segmentation
Monthly Berlin CPI for the food sub-index	Euronet or Statistisches Bundesamt (Destatis)	API	Macroeconomic control
Organic retail store locations, Wochenmarkt locations,	OpenStreetMap; Wochenmarkt Deutschland	Overpass API; wochenmarkt-deutschland.de/wochenmarkt-berlin/	H4
Public holidays and major Berlin events calendar	Destatis	Manual	Closure effects, patterns

6.4 Proposed Testing Workflow

1. Proxy Construction: Assemble Google Trends weekly index for all four search terms using the proposed weights. Normalize to 0–100 scale. Aggregate to weekly (Mon–Sun) cadence covering January 2021 to present.
2. Public Data Assembly: Compile neighborhood-level sociodemographic variables from sources. Map organic retail and market density to Berlin's neighborhood zones. Collect weather data for the same weekly periods as the Trends index.
3. AgriCom Data Integration: Upon receiving sales and price data from AgriCom, we then merge with the proxy index on the weekly identifier and product. Validate alignment of date ranges and units of measurement.

4. Correlation and Calibration (H1, H2): Run correlations between the Google Trends index and weekly sales volumes per product. Test composite index performance against equal-weight benchmark. Estimate a scaling coefficient to convert the normalized index into approximate demand units.
5. Elasticity Estimation (H3): Estimate regression of weekly demand on own-price per product, controlling for the season and promotion flags (if provided). Compare elasticity estimates across products.
6. Segmentation Model (H4): Run a cross-sectional regression of organic sales intensity (per district) on the assembled sociodemographic proxies. Interpret the coefficients as segment indicators and then validate against our established five consumer profiles.
7. Weather and Seasonality (H5): Add the weather variables to the time-series model. Test significance of temperature and sunshine hours for each product. Compare model fit with and without weather controls.

References:

Aertsens, J., Verbeke, W., Mondelaers, K., & Van Huylenbroeck, G. (2009). Personal determinants of organic food consumption: A review. British Food Journal, 111(10), 1140–1167.

Boone, T., Ganeshan, R., Hicks, R. L., & Sanders, N. R. (2018). Can google trends improve your sales forecast?. *Production and Operations Management*, *27*(10), 1770-1774.

Boekhout, R. (2022, December 21). Germany: No land, no problem – vertical farming on urban waterways. VerticalFarmDaily.com. https://www.verticalfarmdaily.com/article/9488862/germany-no-land-no-problem-vertical-farming-on-urban-waterways/

Bund Ökologische Lebensmittelwirtschaft (BÖLW). (2025). *Die Bio-Branche 2025*. https://www.boelw.de/service/mediathek/broschuere/die-bio-branche-2025/

Choi, H., & Varian, H. (2012). Predicting the present with Google Trends. *Economic record*, *88*, 2-9.

Goel, S., Hofman, J. M., Lahaie, S., Pennock, D. M., & Watts, D. J. (2010). Predicting consumer behavior with Web search. *Proceedings of the National academy of sciences*, *107*(41), 17486-17490.

Hernández-Cruz, X., Villalobos, J. R., Runger, G., & Neal, G. (2023). Building an intelligent system to identify trends in agricultural markets. *Journal of Cleaner Production*, *425*, 138956.

Hinrichs, C. C. (2000). Embeddedness and local food systems: Notes on two types of direct agricultural market. Journal of Rural Studies, 16(3), 295–303.

Huang, K. S., & Lin, B.-H. (2007). Estimation of food demand and nutrient elasticities from household survey data. United States Department of Agriculture, Economic Research Service, Technical Bulletin No. 1887.

Hughner, R. S., McDonagh, P., Prothero, A., Shultz, C. J., & Stanton, J. (2007). Who are organic food consumers? A compilation and review of why people purchase organic food. Journal of Consumer Behaviour, 6(2–3), 94–110.

Hu, T., Al Mamun, A., Reza, M. N. H., Wu, M., & Yang, Q. (2024). Examining consumers' willingness to pay premium price for organic food. *Humanities and Social Sciences Communications*, *11*(1), 1–15.

Janssen, M., & Hamm, U. (2012). Product labelling in the market for organic food: Consumer preferences and willingness-to-pay for different organic certification logos. Food Quality and Preference, 25(1), 9–22.

Murray, Kyle & Muro, Fabrizio & Finn, Adam & Popkowski Leszczyc, Peter. (2010). The Effect of weather on consumer spending. Journal of Retailing and Consumer Services. 17. 512-520. 10.1016/j.jretconser.2010.08.006.

Reisch, L., Eberle, U., & Lorek, S. (2013). Sustainable food consumption: An overview of contemporary issues and policies. Sustainability: Science, Practice and Policy, 9(2), 7–25.

Richards, T. J., Bontemps, C., Bouamra-Mechemache, Z., & Réquillart, V. (2021). Retail market power in a shopping basket model of supermarket competition. Journal of Industrial Economics, 66(4), 883–924.

Thünen Institute. (2024). *Current trends in the German organic sector*. https://www.thuenen.de/en/thuenen-topics/organic-farming/aktuelle-trends-der-deutschen-oekobranche

Van Dijk, A., van Heerde, H. J., Leeflang, P. S. H., & Wittink, D. R. (2004). Similarity-based spatial methods to estimate shelf-based substitution patterns. Quantitative Marketing and Economics, 2(3), 257–287.

Yildiz, E. (2009). Die weltoffene Stadt: Wie Migration Globalisierung zum urbanen Alltag macht. transcript Verlag.